

A PROOF OF THE TWO-DIMENSIONAL GAUSSIAN-MOMENTS AND NULLCONE CONJECTURES, FORMALLY VERIFIED

ELIOTT CASSIDY

ABSTRACT. Let E be the Wick (Gaussian) expectation on $\mathbb{C}[Z, W]$ determined by $E[Z^a W^b] = a! [a = b]$, the moment functional of a single complex Gaussian. The *Gaussian-Moments Conjecture* and the *Nullcone Conjecture* of Derksen, van den Essen and Zhao assert, respectively, that the kernel of the Gaussian moment functional is a Mathieu-Zhao space, and that a moment-null polynomial has one-sided charge. The former implies the Jacobian Conjecture but fails in higher dimension, so dimension two is the sharp surviving case. We prove both conjectures in dimension two: if $E(P^m) = 0$ for all $m \geq 1$ then $E(Q P^m) = 0$ for all large m , and every P with $E(P^m) = 0$ for all m has strictly one-sided charge. The proof deduces the moment statement from the nullcone statement and reduces the residual analytic content to the one-variable Duistermaat-van der Kallen constant-term theorem. We give a self-contained, valuation-free proof of the latter through a Galois orbit-product on the splitting field of $X^M - tR$, fed by the Weierstrass small-root product $\Pi = ct$, whose analytic heart $\partial_t h(0, t) = 0$ is proved inside a single power-series frame. The entire argument has been formalized in the Lean 4 proof assistant on top of `mathlib`; the top-level theorem is machine-checked and depends only on the three foundational axioms of that library.

1. INTRODUCTION

1.1. The conjecture. Fix the polynomial ring $\mathbb{C}[Z, W]$ in two variables and regard $W = \bar{Z}$ as the conjugate of a single complex Gaussian variable Z . The associated *Wick*, or *Gaussian*, expectation is the linear functional

$$E: \mathbb{C}[Z, W] \longrightarrow \mathbb{C}, \quad E[Z^a W^b] = a! [a = b], \quad (1)$$

where $[a = b]$ is 1 if $a = b$ and 0 otherwise. Equivalently, writing $Z = x + iy$, $W = x - iy$, E is integration against the standard planar Gaussian, and (1) is the classical Wick formula for its moments. The functional E annihilates every monomial off the diagonal, i.e. every monomial of nonzero *charge*

$$\text{ch}(Z^a W^b) = a - b.$$

A linear subspace $\mathcal{M}$ of a commutative $\mathbb{C}$ -algebra A is a *Mathieu-Zhao space* (Zhao [4]; originally *Mathieu subspace*) if, whenever $a \in A$ satisfies $a^m \in \mathcal{M}$ for all $m \geq 1$, one has $ba^m \in \mathcal{M}$ for all sufficiently large m and every $b \in A$. These spaces isolate the algebraic content of a family of conjectures descending from the Jacobian Conjecture [7]: Mathieu's conjecture on integrals of G -finite functions [3], Zhao's Image Conjecture and its Hessian-nilpotent form [6, 5], and their common Gaussian shadow.

Derksen, van den Essen and Zhao [1] formulated, and connected to the Jacobian Conjecture, two conjectures about the Gaussian moment functional. The first asks that its kernel be a Mathieu-Zhao space.

Date: July 23, 2026.

2020 Mathematics Subject Classification. 14R15, 13F20, 68V20, 12E05.

Key words and phrases. Gaussian-Moments Conjecture, Mathieu-Zhao space, Jacobian Conjecture, Duistermaat-van der Kallen theorem, constant terms, Weierstrass preparation, formalization, Lean.

Conjecture 1.1 (Gaussian–Moments Conjecture, [1]). *The kernel of the Gaussian moment expectation is a Mathieu–Zhao space of the ambient polynomial ring.*

They proved that Conjecture 1.1 implies the Jacobian Conjecture, and observed that it does not extend to arbitrary dimension: a counterexample to the unrestricted moment–vanishing statement shows that the Gaussian–Moments phenomenon fails once enough Gaussian variables are present. In the two–variable ring $\mathbb{C}[Z, W]$, which realizes a single complex Gaussian Z with $W = \bar{Z}$ via (1), the conjecture survives; this is the sharp case, and it is our first theorem.

Theorem 1.2 (Gaussian–Moments Conjecture in dimension two). *Let $P, Q \in \mathbb{C}[Z, W]$. If $E(P^m) = 0$ for all $m \geq 1$, then there is N such that $E(Q P^m) = 0$ for all $m \geq N$.*

The companion *Nullcone Conjecture* of [1] makes the underlying instability structural: a polynomial all of whose central Gaussian moments vanish should lie in the nullcone of the relevant torus action, i.e. be destabilized by a one–parameter subgroup. In dimension two the torus is the single hyperbolic $\mathbb{C}^\times$ acting with weight $\text{ch}(s) = s_0 - s_1$ on $Z^{s_0}W^{s_1}$, and its nullcone consists precisely of the strictly one–sided polynomials. Our second theorem is this two–dimensional Nullcone Conjecture.

Theorem 1.3 (Nullcone Conjecture in dimension two). *Let $P \in \mathbb{C}[Z, W]$ and write $\text{supp } P \subset \mathbb{N}^2$ for its exponent support. If $E(P^m) = 0$ for all $m \geq 1$, then P has one–sided charge: either $\text{ch}(s) \geq 1$ for every $s \in \text{supp } P$, or $\text{ch}(s) \leq -1$ for every $s \in \text{supp } P$.*

We abbreviate Theorems 1.2 and 1.3 by GMC(2) and NC(2). Theorem 1.3 is the structural heart of Theorem 1.2: it says that the only way for all central powers of P to be moment–null is for P to be strictly one–sided (no monomial can have charge 0, and all charges must agree in sign). The passage $\text{NC}(2) \Rightarrow \text{GMC}(2)$ is elementary (Section 2); the content is Theorem 1.3 itself.

1.2. Method. The proof of Theorem 1.3 confronts one, and only one, genuinely analytic input: the one–variable *Duistermaat–van der Kallen theorem* on constant terms in powers of a Laurent polynomial [2]. In the exact form we need it reads:

Theorem 1.4 (Duistermaat–van der Kallen, finite–support form). *Let $f(z) = \sum_i c_i z^{q_i}$ be a Laurent polynomial with distinct exponents $q_i \in \mathbb{Z}$ and nonzero coefficients $c_i \in \mathbb{C}$, whose support straddles zero: either 0 occurs among the q_i , or both a negative and a positive exponent occur. Then there is $m \geq 1$ with $\text{CT}(f^m) \neq 0$, where CT denotes the constant Laurent coefficient.*

The classical theorem of Duistermaat and van der Kallen [2] classifies the Laurent polynomials all of whose positive powers have vanishing constant term; Theorem 1.4 is the contrapositive in the two–sided case, together with the trivial zero–charge case. We do not import the analytic proof of [2]. Instead we give a self–contained algebraic proof (Sections 4–6) that runs as follows. Clearing denominators turns f into a genuine polynomial and reduces the vanishing of all $\text{CT}(f^m)$ to the vanishing of the central coefficients $D_m = [X^{Mm}]R(X)^m$ of a single polynomial R ; these are precisely the coefficients governing the small roots of

$$\Phi = X^M - t R(X) \in \mathbb{C}(t)[X].$$

Over the splitting field of Φ , a transitive Galois action and a Vieta identity force a contradiction *as soon as* the product of the M small roots equals ct with $c \neq 0$ (Section 5). That product identity is exactly the Weierstrass small–root product $\Pi = ct$, whose only nontrivial ingredient is the scalar analytic identity $\partial_t h(0, t) = 0$ under $D_m \equiv 0$, proved inside the single power–series frame $(\mathbb{C}((x)))[[t]]$ (Section 6). No extension of a valuation to an algebraic closure, and no Puiseux expansion, is required.

1.3. Formalization. Every step above — Theorems 1.2, 1.3, 1.4, the univariate reduction, the orbit–product contradiction, and the Weierstrass analysis — has been formalized in the Lean 4 proof assistant on top of `mathlib` [8]. The top–level statement is

```
theorem GMC2.gmc2 (P Q : MvPolynomial (Fin 2) ℂ)
  (hnull : ∀ m, 1 ≤ m → E (P ^ m) = 0) :
  ∃ N, ∀ m ≥ N, E (Q * P ^ m) = 0
```

and a kernel audit reports

```
#print axioms GMC2.gmc2 = [propxt, Classical.choice, Quot.sound],
```

the three standard axioms of `mathlib`: there is no `sorry`, no appeal to `native_decide`, and no additional axiom anywhere in the transitive dependency. In particular the one–variable input, Theorem 1.4, is itself formalized rather than assumed. The proof is described module by module in Section 7.

1.4. Organization. Section 2 records the functional E and the elementary reduction $\text{NC}(2) \Rightarrow \text{GMC}(2)$. Section 3 reduces $\text{NC}(2)$ to Theorem 1.4. Section 4 reduces Theorem 1.4 to a single polynomial statement (the *small–root packet product*). Section 5 closes that statement by a Galois orbit–product. Section 6 supplies the Weierstrass small–root product and its analytic core. Section 7 documents the Lean formalization and its verification. Section 8 collects remarks on sharpness and on the Jacobian Conjecture.

2. THE WICK FUNCTIONAL AND THE CHARGE REDUCTION

Write monomials of $\mathbb{C}[Z, W]$ as $Z^{s_0}W^{s_1}$ with $s = (s_0, s_1) \in \mathbb{N}^2$, and set

$$\text{wt}(s) = \begin{cases} s_0! & s_0 = s_1, \\ 0 & \text{otherwise,} \end{cases} \quad E(P) = \sum_{s \in \text{supp } P} P_s \text{wt}(s),$$

where P_s is the coefficient of $Z^{s_0}W^{s_1}$ in P . Thus E agrees with (1) and annihilates every monomial of nonzero charge $\text{ch}(s) = s_0 - s_1$.

Lemma 2.1. *If $P, R \in \mathbb{C}[Z, W]$ have the property that every monomial of P and every monomial of R has charge ≥ 1 (respectively ≤ -1), then $E(R) = 0$; more generally $E(PR) = 0$ whenever every monomial of PR has nonzero charge.*

Proof. Charge is additive under multiplication, $\text{ch}(s + t) = \text{ch}(s) + \text{ch}(t)$, so every monomial of a product of strictly one–sided polynomials has charge of the same strict sign, hence nonzero; and wt vanishes off charge 0. $\square$

Proposition 2.2 ($\text{NC}(2) \Rightarrow \text{GMC}(2)$). *Assume Theorem 1.3. Then Theorem 1.2 holds; explicitly, if $E(P^m) = 0$ for all $m \geq 1$ then $E(Q P^m) = 0$ for all $m \geq \deg_W Q + 1$ (in the case of positive charge; symmetrically for negative charge).*

Proof. By Theorem 1.3, P is strictly one–sided; say every monomial of P has charge ≥ 1 (the other case is symmetric under $Z \leftrightarrow W$). Every monomial of P^m then has charge $\geq m$. A monomial $Z^a W^b$ of Q has $b \leq \deg_W Q$, so its charge is $\geq -\deg_W Q$; hence a monomial of $Q P^m$ has charge $\geq m - \deg_W Q \geq 1$ once $m \geq \deg_W Q + 1$. By Lemma 2.1, $E(Q P^m) = 0$ for all such m . $\square$

The threshold $N = \deg_W Q + 1$ is explicit and uniform: the whole difficulty is concentrated in Theorem 1.3, to which we now turn.

3. REDUCTION OF $\text{NC}(2)$ TO THE CONSTANT-TERM THEOREM

Write $P = \sum_{i=1}^k c_i Z^{a_i} W^{b_i}$ with $c_i \in \mathbb{C}^\times$ and distinct exponent pairs, and let $q_i = a_i - b_i$ be the charges. Assume $E(P^m) = 0$ for all $m \geq 1$ and, for contradiction, that P is not one-sided, so that 0 lies in the convex hull of $\{q_1, \dots, q_k\}$. The proof extracts an extremal one-variable face, applies Theorem 1.4 to it to obtain a single nonzero constant term, and amplifies that scalar into a nonzero moment by a characteristic- p Frobenius argument.

The moment expansion. Since E annihilates every nonzero charge, only charge-balanced multi-indices survive in a power. For $r \in \mathbb{N}^k$ put $|r| = \sum_i r_i$, $c^r = \prod_i c_i^{r_i}$, and, when $\sum_i q_i r_i = 0$, let $A(r) = \sum_i a_i r_i = \sum_i b_i r_i$ (the two agree by balance) be the common diagonal degree. Then

$$E(P^m) = M_m(c) := \sum_{\substack{r \in \mathbb{N}^k, |r|=m \\ \sum_i q_i r_i = 0}} \binom{m}{r} A(r)! c^r, \quad \binom{m}{r} = \frac{m!}{\prod_i r_i!}. \quad (2)$$

The distinguishing feature of dimension two is the single monotone factorial $A(r)!$ on each balanced channel. Its presence is exactly why the vanishing of $E(P^m)$ is *not* the vanishing of the plain constant term of the associated Laurent polynomial $f(z) = \sum_i c_i z^{q_i}$,

$$\text{CT}(f^m) = \mathcal{C}_m(q)(c), \quad \mathcal{C}_m(q) = \sum_{\substack{r \in \mathbb{N}^k, |r|=m \\ \sum_i q_i r_i = 0}} \binom{m}{r} X^r \in \mathbb{Q}[X_1, \dots, X_k], \quad (3)$$

and it is the crux of the two-dimensional case. We control it through the lowest balanced face.

The lowest balanced face. Consider the linear program

$$\delta = \min \left\{ \sum_i a_i x_i : x \geq 0, \sum_i x_i = 1, \sum_i q_i x_i = 0 \right\}, \quad (4)$$

feasible precisely because $0 \in \text{conv}\{q_i\}$, with value $\delta \in \mathbb{Q}$ attained. Linear programming duality supplies a multiplier $\lambda \in \mathbb{Q}$ with

$$a_i - \lambda q_i \geq \delta \quad (1 \leq i \leq k). \quad (5)$$

Let $F = \{i : a_i - \lambda q_i = \delta\}$ be the tight set. An optimal vertex of (4) is supported on F , so the face charges still straddle zero:

$$0 \in \text{conv}\{q_i : i \in F\}, \quad f_F(z) = \sum_{i \in F} c_i z^{q_i}. \quad (6)$$

Since $a_i = \delta + \lambda q_i$ on F , every balanced channel s supported on F has the *same* diagonal degree

$$A(s) = \sum_{i \in F} a_i s_i = \delta |s| + \lambda \sum_{i \in F} q_i s_i = \delta |s|. \quad (7)$$

The seed. The face polynomial f_F has charges straddling zero, so Theorem 1.4 applies to it: there is $m_0 \geq 1$ with

$$Q := \text{CT}(f_F^{m_0}) = \mathcal{C}_{m_0}(q_F)(c_F) \neq 0, \quad (8)$$

q_F, c_F denoting charges and coefficients on F . Every channel contributing to (8) is balanced, of length m_0 , and supported on F ; by (7) all share the diagonal degree $A_0 := \delta m_0 = A(s) \in \mathbb{N}$. (When F contains a zero charge the seed is immediate, $Q = c_{i_0}$ at $m_0 = 1$; otherwise it is the substance of Theorem 1.4.)

Frobenius amplification. Each M_m of (2) is a polynomial in $\mathbb{Z}[X_1, \dots, X_k]$ evaluated at c , and Q is the value at c of an integer polynomial that does not vanish. Hence the finitely generated $\mathbb{Z}$ -algebra $\mathbb{Z}[c_1, \dots, c_k] \subset \mathbb{C}$ has a maximal ideal avoiding Q , with finite residue field $\mathbb{F}$ of some

prime characteristic p ; let $\varphi: \mathbb{Z}[c_1, \dots, c_k] \rightarrow \mathbb{F}$ be the quotient, $\bar{c}_i = \varphi(c_i)$ and $\bar{Q} = \varphi(Q) \neq 0$. Every identity $M_m(c) = 0$ maps to $M_m(\bar{c}) = 0$ in $\mathbb{F}$.

Take $m = pm_0$. By (5) every balanced channel r of length pm_0 has

$$A(r) = \delta(pm_0) + \sum_i (a_i - \lambda q_i - \delta) r_i \geq pA_0, \quad (9)$$

with equality iff r is supported on F . The minimal factorial in (2) at $m = pm_0$ is thus $(pA_0)!$, and

$$N := \frac{M_{pm_0}}{(pA_0)!} = \sum_{\substack{|r|=pm_0 \\ \sum_i q_i r_i = 0}} \binom{pm_0}{r} \frac{A(r)!}{(pA_0)!} X^r \in \mathbb{Z}[X_1, \dots, X_k]$$

is an *integer* polynomial (each $A(r)!/(pA_0)! \in \mathbb{Z}$ by (9)). From $M_{pm_0}(c) = E(P^{pm_0}) = 0$ we get $N(c) = 0$, hence $N(\bar{c}) = 0$ in $\mathbb{F}$. We evaluate $N(\bar{c})$ a second way, term by term modulo p :

- (1) if r is not componentwise divisible by p , then $p \mid \binom{pm_0}{r}$ (Kummer's theorem; equivalently the Frobenius vanishing of the multinomial), and the term is 0 in $\mathbb{F}$;
- (2) if $r = ps$ with s balanced but not supported on F , then $A(r) = pA(s) \geq p(A_0+1)$ by (9), so $p \mid A(r)!/(pA_0)!$ and the term is 0 in $\mathbb{F}$;
- (3) if $r = ps$ with s balanced and supported on F , then $A(r) = pA_0$ and $A(r)!/(pA_0)! = 1$.

Only the third layer survives, so, using the multinomial Lucas congruence $\binom{pm_0}{ps} \equiv \binom{m_0}{s} \pmod{p}$ and then the Frobenius endomorphism $x \mapsto x^p$ of $\mathbb{F}$ (which fixes the integers $\binom{m_0}{s} \in \mathbb{F}_p$),

$$N(\bar{c}) = \sum_{\substack{|s|=m_0, \sum_i q_i s_i = 0 \\ \text{supp } s \subseteq F}} \binom{pm_0}{ps} \bar{c}^{ps} = \sum_s \binom{m_0}{s} \bar{c}^{ps} = \left(\sum_s \binom{m_0}{s} \bar{c}^s \right)^p = \bar{Q}^p$$

in $\mathbb{F}$. Thus $\bar{Q}^p = N(\bar{c}) = 0$, forcing $\bar{Q} = 0$ and contradicting $\bar{Q} \neq 0$. Therefore $0 \notin \text{conv}\{q_i\}$, i.e. P is one-sided, proving Theorem 1.3. $\square$

Remark 3.1. The mechanism is finite and characteristic- p ; the analytic transcendence of the problem enters *only* through the seed (8), i.e. Theorem 1.4 applied to the one-variable face. In the formalization the rational face F , the height datum $A_0 = \delta m_0$ with the floor (9) and the off-face gap of case (2) are packaged in `GMC2NC2 (HeightWitnessSupplier, discharged unconditionally, and normalized.specialization.contradiction)`; the three-case Frobenius collapse to $\bar{Q}^p$ is `GMC2FrobeniusResidue/GMC2NormalizedResidue`; and the specialization φ is the integral “let the quotient choose the prime” descent of `GMC2IntegralFaceSeedDescent`. The seed itself is $\mathcal{C}_{m_0}(q_F)(c_F) \neq 0$, supplied by `DvdK1` on the face (Sections 4–6).

4. THE UNIVARIATE REDUCTION: FROM CONSTANT TERMS TO A SMALL-ROOT PACKET

Fix injective q straddling zero with no zero charge, and coefficients $c_i \neq 0$. Let $M = -\min_i q_i > 0$ and form the genuine polynomial

$$R(X) = \sum_i c_i X^{q_i+M} \in \mathbb{C}[X].$$

The shift is chosen so that $R(0) = c_{i_0} \neq 0$ (the unique minimal charge) and $M < \deg R$ (the unique maximal charge is strictly positive). A direct expansion identifies the constant-term relations (3) with central coefficients of powers of R :

$$[X^{Mm}] R(X)^m = \mathcal{C}_m(q)(c) \quad (m \geq 0). \quad (10)$$

Thus assuming $\mathcal{C}_m(q)(c) = 0$ for all $m \geq 1$ is the same as assuming $D_m := [X^{Mm}] R^m = 0$ for all $m \geq 1$.

These central coefficients are the moments of the small roots of

$$\Phi = X^M - tR(X) \in \mathbb{C}(t)[X], \quad (11)$$

a polynomial of degree $N = \deg R > M$ in X over the rational function field $\mathbb{C}(t)$. Modulo t it degenerates to X^M , so exactly M of its N roots are “small” (they tend to 0 as $t \rightarrow 0$); their product carries a t -adic valuation equal to 1 when the moments D_m vanish. The whole of Theorem 1.4 reduces to the following single statement about Φ .

Proposition 4.1 (Small-root packet product). *Let $R \in \mathbb{C}[X]$ with $1 \leq M < \deg R$ and $R(0) \neq 0$, and suppose $D_m = [X^{Mm}]R^m = 0$ for all $m \geq 1$. Then, in the splitting field L of $\Phi = X^M - tR$ over $\mathbb{C}(t)$, there is a finite set S of roots of Φ whose product equals ct for some $c \in \mathbb{C}^\times$:*

$$\prod_{\beta \in S} \beta = ct \in \mathbb{C}(t) \subset L.$$

Granting Proposition 4.1, Theorem 1.4 follows: were $D_m = 0$ for all $m \geq 1$, the packet product would be ct , and Section 5 shows this is impossible; hence some $D_m \neq 0$, i.e. some $\mathcal{C}_m(q)(c) \neq 0$.

5. THE GALOIS ORBIT-PRODUCT CONTRADICTION

We first record why a packet product equal to ct is impossible, independently of *how* the packet arises. This is the algebraic core, an orbit-product identity for the Galois action on the roots.

Lemma 5.1. *$\Phi = X^M - tR$ is irreducible over $\mathbb{C}(t)$ whenever $1 \leq M$ and $R(0) \neq 0$.*

Sketch. Φ is linear in t with content 1; the coefficients X^M and $-R$ of t^0, t^1 are coprime in $\mathbb{C}[X]$ because $R(0) \neq 0$ (so $X \nmid R$). Hence Φ is irreducible in $\mathbb{C}[X][t] = \mathbb{C}[t][X]$, and by Gauss’s lemma over the fraction field $\mathbb{C}(t)$. (In the formalization this is carried out through the Bivariate swap $\mathbb{C}[t][X] \cong \mathbb{C}[X][t]$ and `mathlib`’s primitive/Gauss machinery.) $\square$

Let L be the splitting field of Φ over $F := \mathbb{C}(t)$ and $G = \text{Gal}(L/F)$. Since Φ is irreducible, G acts transitively on the set Ω of its roots. The key point is that the orbit-product identity below holds for an *arbitrary* G -stable factor of the product — one need not know that S is the “valuation-positive” packet, only that its product is G -fixed.

Lemma 5.2 (Orbit-product). *Let G act transitively on a finite set Ω with an equivariant embedding $f: \Omega \rightarrow L$, and let $S \subseteq \Omega$ be any subset such that $\pi := \prod_{\beta \in S} f(\beta)$ is G -fixed. Then for a basepoint $x \in \Omega$,*

$$\pi^{|G|} = \left(\prod_{\alpha \in \Omega} f(\alpha) \right)^{|S| \cdot |\text{Stab}_G(x)|}.$$

Proof. Applying every $g \in G$ and multiplying, $\prod_{g \in G} g \cdot \pi = \pi^{|G|}$ since π is fixed. On the other hand $\prod_{g \in G} g \cdot \pi = \prod_{\beta \in S} \prod_{g \in G} f(g\beta)$, and by transitivity $\prod_{g \in G} f(g\beta) = \left(\prod_{\alpha \in \Omega} f(\alpha) \right)^{|\text{Stab}_G(x)|}$ for every β . Taking the product over the $|S|$ elements of S gives the claim. $\square$

Proposition 5.3 (No packet has product ct). *With $F = \mathbb{C}(t)$, L, G as above, suppose $S \subseteq \Omega$ has G -fixed product $\prod_{\beta \in S} \beta = ct$ with $c \in \mathbb{C}^\times$. Then a contradiction ensues.*

Proof. Because $ct \in F$ lies in the base field, it is automatically G -fixed, so Lemma 5.2 applies with $\pi = ct$. The full product of the roots is, by Vieta on (11),

$$\prod_{\alpha \in \Omega} \alpha = (-1)^N \frac{[X^0]\Phi}{\text{lc}_X \Phi} = (-1)^N \frac{-tR(0)}{-t \text{lc}_X R} = (-1)^N \frac{R(0)}{\text{lc}_X R} =: d \in \mathbb{C}^\times,$$

a nonzero *constant*: the factor t cancels between the constant and leading coefficients of Φ . Substituting into Lemma 5.2,

$$(ct)^{|G|} = d^{|S| \cdot |\text{Stab}_G(x)|} \quad \text{in } F = \mathbb{C}(t).$$

The left side has t -adic valuation $|G| \geq 1$; the right side is a nonzero constant of valuation 0. A degree- $|G|$ monomial in t cannot equal a constant, so this is absurd. $\square$

Proposition 5.3 is what makes the reduction *valuation-free*. Nothing about the size of the roots is used; the only inputs are transitivity (Lemma 5.1), Vieta, and the elementary fact that a monomial is not a constant. It remains to produce a packet with product ct , i.e. to prove Proposition 4.1. This is where the Weierstrass analysis enters, and it is the only place any analysis is used.

6. THE WEIERSTRASS SMALL-ROOT PRODUCT AND ITS ANALYTIC CORE

Regard Φ as a polynomial in X over the complete local ring $A = \mathbb{C}[[t]]$:

$$\Phi = X^M - tR(X) \in A[X] \subset A[[X]].$$

Its reduction mod t is X^M , so Weierstrass preparation over $A[[X]]$ yields a factorization

$$\Phi = P \cdot h, \quad P \in A[X] \text{ monic of degree } M, \quad P \equiv X^M \pmod{t}, \quad h \in A[[X]]^\times. \quad (12)$$

P is the *distinguished* (small-root) factor; its M roots are the small roots of Φ , and the small-root product is $\Pi = (-1)^M P.\text{coeff}_0$.

6.1. From power series to a polynomial factorization. The factorization (12) lives in the power-series ring; to feed the packet into Section 5 we need P to divide Φ as *polynomials*, so that the roots of P are genuinely roots of Φ in L .

Lemma 6.1. *P divides Φ in the polynomial ring $A[X]$.*

Proof. Since P is monic, ordinary polynomial division gives $\Phi = P \cdot Q + r$ with $\deg_X r < M$. Coercing to $A[[X]]$ and comparing with (12), both $(h, 0)$ and (Q, r) are Weierstrass divisions of Φ by P ; since $P \equiv X^M \pmod{t}$ makes P a valid Weierstrass divisor over the Hausdorff ring A , uniqueness of Weierstrass division forces the remainder r to vanish. Hence $P \mid \Phi$ in $A[X]$. $\square$

Mapping $A = \mathbb{C}[[t]] \hookrightarrow \mathbb{C}((t))$ and then into $\Omega = \overline{\mathbb{C}((t))}$, and using that the polynomial (11) over $\mathbb{C}(t)$ coincides with its image over $\mathbb{C}((t))$, one obtains $P \mid \Phi$ over Ω . The splitting field L of Φ over $\mathbb{C}(t)$ embeds $\mathbb{C}(t)$ -linearly into Ω by an embedding $\psi: L \hookrightarrow \Omega$ (both fields split Φ), and the packet

$$S = \{ \beta \in L : \Phi(\beta) = 0 \text{ and } \psi(\beta) \text{ is a root of } P \}$$

has, by Vieta on the monic P ,

$$\psi\left(\prod_{\beta \in S} \beta\right) = \prod (\text{roots of } P) = (-1)^M P.\text{coeff}_0.$$

Since ψ is injective and fixes $\mathbb{C}(t)$, the identity descends to L : $\prod_{\beta \in S} \beta = (-1)^M P.\text{coeff}_0$ once the latter is shown to be a base-field element. This is the content of the next, and final, computation.

6.2. The scalar identity $\Pi = ct$. Taking the X^0 -coefficient of (12) gives, with $h(0, t) := [X^0]h \in A^\times$ and $r_0 = R(0)$,

$$P.\text{coeff}_0 \cdot h(0, t) = [X^0]\Phi = -tr_0, \quad (13)$$

so $\Pi = (-1)^M P.\text{coeff}_0$ equals ct with $c = (-1)^{M+1}r_0$ if and only if $h(0, t) = 1$. The unit $h(0, t) \in \mathbb{C}[[t]]$ is normalized by $h(0, 0) = 1$ (from $P \equiv X^M \equiv \Phi \pmod{t}$ and cancellation of X^M in the domain $\mathbb{C}[[X]]$). It remains to show that its t -derivative vanishes:

$$\partial_t h(0, t) = 0 \quad \text{when } D_m = 0 \text{ for all } m \geq 1. \quad (14)$$

Over a field of characteristic 0, (14) together with $h(0, 0) = 1$ forces $h(0, t) = 1$, hence $\Pi = ct$, hence Proposition 4.1.

Identity (14) is the sole analytic ingredient of the entire paper, and it is a one-line logarithmic-derivative computation once the objects are placed in a *single* completion. Work in

$$\mathcal{A} = (\mathbb{C}((x)))[[t]],$$

the t -power series over the Laurent field $\mathbb{C}((x))$. Here $\Phi = x^M - tR(x)$, P and h all map in and are *units* (their t -constant terms are $x^M, x^M, 1$, all nonzero in the field $\mathbb{C}((x))$), so R/Φ , $\partial_t P/P$, $\partial_t h/h$ are honest elements of $\mathcal{A}$. Differentiating $\Phi = P \cdot h$ in t and dividing by Φ ,

$$-\frac{R}{\Phi} = \frac{\partial_t P}{P} + \frac{\partial_t h}{h}.$$

Apply the $\mathbb{C}[[t]]$ -linear functional $[x^0]$ (the x^0 -Laurent coefficient, a ring homomorphism on the “disk” subring of series with nonnegative x -support). Three evaluations finish the argument:

- (i) $[x^0](\partial_t P/P) = 0$: P is monic of x -degree M and $\partial_t P$ has x -degree $< M$, so $\partial_t P/P$ has strictly negative x -degree and no x^0 term (a reversed-polynomial degree count);
- (ii) $[x^0](-R/\Phi) = -\sum_{m \geq 1} D_m t^{m-1}$: expanding $x^M/\Phi = (1 - tR/x^M)^{-1}$ as a geometric series identifies the x^0 coefficient with the moment generating function $\sum_m D_m t^m$;
- (iii) $[x^0](\partial_t h/h) = \partial_t h(0, t)/h(0, t)$: h and $\partial_t h$ have nonnegative x -support, so on that subring $[x^0]$ is the ring homomorphism $h \mapsto h(0, t)$, which commutes with ∂_t .

Combining, $-\sum_{m \geq 1} D_m t^{m-1} = 0 + \partial_t h(0, t)/h(0, t)$. Under $D_m \equiv 0$ the left side is 0, so $\partial_t h(0, t) = 0$, proving (14) and completing the proof of Proposition 4.1, hence of Theorems 1.4, 1.3, and 1.2. $\square$

Remark 6.2. The passage between the “space” completion $(\mathbb{C}((x)))[[t]]$ used for (14) and the “deformation” completion $\mathbb{C}[[t]][X]$ used for Lemma 6.1 is a transposition of the two power-series variables; both are algebraic and neither requires extending a valuation to the algebraic closure of $\mathbb{C}((t))$. This is the decisive simplification over the Puiseux/Newton-polygon route: the orbit-product contradiction (Proposition 5.3) accepts *any* packet whose product is ct , so the packet may be defined algebraically as the roots of the Weierstrass factor P rather than by valuation.

7. THE LEAN FORMALIZATION

The proof is formalized in Lean 4 against `mathlib` [8]. We summarize the module structure; the correspondence to the sections above is exact.

- **The functional and the charge reduction.** `E`, `wt`, `charge`, `NC2`, and the reduction `gmc2.of_nc2` (Proposition 2.2) are in `GMC2Reduction`. The threshold $N = \deg_W Q + 1$ is explicit.

- **NC(2) $\Leftarrow$ DvdK(1).** The lowest-balanced-face and Frobenius argument of Section 3 is `GMC2NC2` (`normalized_specialization_contradiction`, `HeightWitnessSupplier`), with the characteristic- p residue computation in `GMC2FrobeniusResidue` and `GMC2NormalizedResidue` and the integral face-seed descent in `GMC2IntegralFaceSeedDescent`. The height witness is discharged in `GMC2HeightWitness`, where the unconditional endpoints `nc2_of_dvdK1` and `gmc2_of_dvdK1` live.
- **The constant-term interface.** `DvdK1` (Theorem 1.4), `ChargesStraddleZero`, and the universal $\mathcal{C}_m(q)$ (3) (`constantTermRelation`) are in `GMC2DvdKInterface`, `GMC2LowestFaceExistence`, `GMC2ConstantTermRelations`; the zero-charge case and the both-signs reduction are in `GMC2DvdKZeroCharge`.
- **Univariate reduction.** The shift $R = \text{shiftedPolynomial}$ and the identity (10) are in `GMC2LaurentShiftCheckA`; `SinglePolyCrux` (Proposition 4.1) and the capstone `gmc2_of_crux` are in `GMC2DvdKUnivariateReduction`.
- **Orbit-product.** Lemma 5.2 and Proposition 5.3 are `GMC2OrbitProduct` and `GMC2Thm2067Wrapper` (`thm2067_contradiction`); irreducibility (Lemma 5.1) is `GMC2PhiIrreducible/GMC2DvdKAssembly`; Vieta ($\prod \alpha = d$) is `GMC2PhiVieta` (`prod_rootSet_Phi`); the char-0 specialization discharging separability and Galois-fixedness is `GMC2Thm2067HSonly` (`thm2067_reduced_to_hS`).
- **The frame bridge.** The valuation-free packet transport of Section 6 — root transport, the packet bijection, and the assembly $\prod_{\beta \in S} \beta = \text{algebraMap}(ct)$ — is `GMC2FrameBridgeRoots`, `GMC2FrameBridgePacket` (`exists_packet_prodeq`, a `Finset.prod_bij`), `GMC2FrameBridge`, `GMC2FrameBridgeAssembly`.
- **Weierstrass and the analytic core.** (12), $P = \text{smallRootFactor}$, and (13) are `GMC2DvdKWeierstrass` (built on `mathlib`'s Weierstrass preparation); Lemma 6.1 is `GMC2FrameBridgeDvd` (`smallRootFactor_dvd_PhiPoly`); the normalization $h(0,0) = 1$ is `GMC2DvdKUnitOrigin`; the log-derivative identity (14) is proved in the unified frame ($\mathbb{C}((x))[[t]]$ across `GMC2DvdKFrame`, `GMC2DvdKTranspose`, `GMC2DvdKHderivAssembly`, `GMC2DvdKTransposeAssembly`; the char-0 “vanishing derivative $\Rightarrow$ constant” step is `GMC2DvdKCharZeroClosing`.
- **Discharge of the crux and the front door.** The wiring over $\Omega = \overline{\mathbb{C}((x))}$ that proves `singlePolyCrux_holds`, and the unconditional theorem `gmc2_unconditional = gmc2_of_crux singlePolyCrux_holds`, are in `GMC2DvdKOmegaWiring`; the documented statement `GMC2.gmc2` is in `GMC2Main`.

Verification. The build target `GMC2Main` compiles the entire proof and is self-contained. The kernel axiom audit

```
#print axioms GMC2.gmc2 = [propext, Classical.choice, Quot.sound]
```

certifies that the theorem uses only `mathlib`'s three foundational axioms: there is no `sorry`, no `native_decide` (hence no unaudited kernel reduction), and no additional axiom. Every module terminates in a `#print axioms` directive with the same result, so the property holds throughout the dependency graph, not merely at the root.

8. CONCLUDING REMARKS

Sharpness. Theorem 1.2 is dimension-optimal: by [1] the Gaussian-Moments phenomenon fails once enough Gaussian variables are present. The obstruction to higher dimensions is genuinely multi-radial: the single hyperbolic charge $\text{ch} = s_0 - s_1$ of $\mathbb{C}[Z, W]$ is replaced by a lattice of charges, and one-sidedness is no longer forced. Theorem 1.3 is the exact statement that survives in dimension two: instability of the moment functional (all powers null) coincides with

algebraic instability (a strict one-sided charge, i.e. a destabilizing one-parameter subgroup of the hyperbolic torus).

The one-variable input. The argument isolates the entire transcendental difficulty of the problem in the single scalar identity (14), a formal logarithmic derivative. This is a constructive counterpart to the analytic proof of Duistermaat and van der Kallen [2], replacing residues and monodromy of algebraic functions by a Weierstrass factorization and a Galois orbit-product, and in particular giving a proof of Theorem 1.4 that is elementary enough to be checked by a proof assistant.

Relation to the Jacobian Conjecture. By [1], the Gaussian-Moments Conjecture implies the Jacobian Conjecture. Since the Gaussian-Moments Conjecture itself fails in higher dimension, this implication cannot be run in full; the two-dimensional case established here is its sharp positive instance.

Availability. The Lean sources build against `mathlib` and reproduce the axiom audit above. The proof was developed and machine-checked in the `TournamentH7` Lean project; the theorem `GMC2.gmc2` and its self-contained build target `GMC2Main` are the canonical entry points.

REFERENCES

- [1] H. Derksen, A. van den Essen, and W. Zhao, *The Gaussian moments conjecture and the Jacobian conjecture*, Israel J. Math. **219** (2017), no. 2, 917–928. [arXiv:1506.05192](#).
- [2] J. J. Duistermaat and W. van der Kallen, *Constant terms in powers of a Laurent polynomial*, Indag. Math. (N.S.) **9** (1998), no. 2, 221–231.
- [3] O. Mathieu, *Some conjectures about invariant theory and their applications*, in *Algèbre non commutative, groupes quantiques et invariants* (Reims, 1995), Sémin. Congr. **2**, Soc. Math. France, 1997, pp. 263–279.
- [4] W. Zhao, *Generalizations of the image conjecture and the Mathieu conjecture*, J. Pure Appl. Algebra **214** (2010), no. 7, 1200–1216.
- [5] W. Zhao, *Hessian nilpotent polynomials and the Jacobian conjecture*, Trans. Amer. Math. Soc. **359** (2007), no. 1, 249–274.
- [6] W. Zhao, *Images of commuting differential operators of order one with constant leading coefficients*, J. Algebra **324** (2010), no. 2, 231–247.
- [7] A. van den Essen, *Polynomial Automorphisms and the Jacobian Conjecture*, Progress in Mathematics **190**, Birkhäuser, Basel, 2000.
- [8] The mathlib Community, *The Lean mathematical library*, in *CPP 2020*, ACM, 2020, pp. 367–381.
Email address: `eliottcassidy2000@gmail.com`